Chapter 4. Inference for Two Populations

Rank-sum test, shift estimation, paired comparisons, scale tests, Kolmogorov–Smirnov

Contents

4.1 Models for two populations

$X_1, \dots, X_m \overset{\text{i.i.d.}}{\sim} F$ and $Y_1, \dots, Y_n \overset{\text{i.i.d.}}{\sim} G$ independent, $N = m + n$. Typical models:

- **Shift:** $G(x) = F(x - \Delta)$, i.e. $Y \overset{d}{=} X + \Delta$.
- **Scale:** $G(x) = F(x/\gamma)$ (common center 0), or location–scale $G(x) = F((x - \Delta)/\gamma)$.
- **General:** $H_0 : F = G$ against $F \neq G$.

Permutation tests require exchangeability under H_0, which holds when H_0 forces $F = G$: the shift model with $\Delta = 0$, the scale model with $\gamma = 1$, or the general null. It fails for "equal locations, different scales." For that problem see Section 4.6.

For normal data the pooled t-test is optimal when variances are equal, and Welch's test, with $T = (\bar{Y} - \bar{X})/\sqrt{s_X^2/m + s_Y^2/n}$ and Satterthwaite degrees of freedom, is approximately valid when they are not.

4.2 The Wilcoxon rank-sum (Mann–Whitney) test

Motivation. Replacing observations by their ranks in the pooled sample makes the permutation distribution independent of the data, so it can be tabulated once. The rank-sum statistic also estimates $P(X < Y)$, which gives it a natural interpretation.

Rank all N observations together. Let $R_1, \dots, R_n$ be the ranks of the Y's.

Definition (Rank-sum and Mann–Whitney statistics)

$W = \sum_{j=1}^{n} R_j$ and $U = \sum_{i=1}^{m} \sum_{j=1}^{n} I(Y_j > X_i)$ (with $\frac{1}{2}$ for ties).

Proposition 4.1

Without ties, $W = U + n(n+1)/2$.

Proof. $R_j = 1 + \#\{k : Y_k < Y_j\} + \#\{i : X_i < Y_j\}$. Summing over j, the middle terms give $0 + 1 + \cdots + (n-1)$. Hence $W = n + \frac{n(n-1)}{2} + U$. ■

Under $H_0 : F = G$ (continuous), the rank vector of the Y's is a uniformly random n-subset of $\{1, \dots, N\}$. So the null distribution of W depends only on (m, n), and the test is distribution-free. It is the permutation test applied to ranks; large W indicates $\Delta > 0$.

Example 4.1

$X = \{1.1, 2.4\}$, $Y = \{2.0, 3.5, 4.2\}$. The pooled ranks of Y are $2, 4, 5$, so $W = 11$ and $U = 5$. Among the $\binom{5}{3} = 10$ rank subsets, only $\{2, 4, 5\}$ and $\{3, 4, 5\}$ have sum ≥ 11, so $P(W \geq 11) = 0.2$.

4.2.1 Large-sample approximation via sampling without replacement

Theorem 4.2

Let $\bar{X}_n$ be the mean of a simple random sample of size n drawn without replacement from a finite population $\{a_1, \dots, a_N\}$ with mean μ and variance $\sigma^2 = \frac{1}{N}\sum(a_k - \mu)^2$. Then

$$E\bar{X}_n = \mu, \qquad \operatorname{Var}\bar{X}_n = \frac{\sigma^2}{n} \cdot \frac{N-n}{N-1}.$$

Proof [Supplement]. Write $n\bar{X}_n = \sum_k Z_k a_k$ with inclusion indicators Z_k, $P(Z_k = 1) = n/N$ and $P(Z_k = Z_l = 1) = \frac{n(n-1)}{N(N-1)}$ for $k \neq l$. So $\operatorname{Var} Z_k = \frac{n}{N}(1 - \frac{n}{N})$ and $\operatorname{Cov}(Z_k, Z_l) = -\frac{n(N-n)}{N^2(N-1)}$. Take $\mu = 0$ without loss, so $\sum_k a_k^2 = N\sigma^2$ and $\sum_{k \neq l} a_k a_l = -N\sigma^2$. Then

$$\operatorname{Var}\Big(\sum Z_k a_k\Big) = N\sigma^2 \frac{n(N-n)}{N^2} + N\sigma^2 \frac{n(N-n)}{N^2(N-1)} = n\sigma^2 \frac{N-n}{N-1}.$$

■

The factor $\frac{N-n}{N-1}$ is the **finite-population correction**. A CLT holds when $\min(n, N-n) \to \infty$ and no single a_k dominates, e.g. $\max_k(a_k - \mu)^2/(\min(n, N-n)\sigma^2) \to 0$ (Hájek). For ranks, $\{a_k\} = \{1, \dots, N\}$ has $\mu = \frac{N+1}{2}$ and $\sigma^2 = \frac{N^2-1}{12}$, so

$$E(W) = \frac{n(N+1)}{2}, \qquad \operatorname{Var}(W) = \frac{mn(N+1)}{12}, \qquad \frac{W - E(W)}{\sqrt{\operatorname{Var}(W)}} \xrightarrow{d} N(0,1).$$

With ties, use the mid-ranks as the finite population: the mean is unchanged, and the variance is recomputed from (4.1) below,

$$\operatorname{Var}(W) = n^2 \cdot \frac{\sigma_R^2}{n} \cdot \frac{N-n}{N-1}, \qquad \sigma_R^2 = \text{population variance of the mid-ranks.} \tag{4.1}$$

The same theorem gives the large-sample permutation distribution of $\bar{Y}$ itself (population = the pooled observations), which is why permutation and normal-theory p-values agree for large samples.

4.2.2 Estimating the shift

In the shift model, $U(\Delta_0) = \#\{(i,j) : Y_j - X_i > \Delta_0\}$ is the Mann–Whitney statistic of X and $Y - \Delta_0$. At the true Δ it has the null distribution, which is symmetric about $mn/2$.

Proposition 4.3

Let $D_{(1)} \le \cdots \le D_{(mn)}$ be the ordered differences $Y_j - X_i$, $k_l = Q_U(\alpha/2)$ and $k_u = mn - k_l + 1$. Then $P(D_{(k_l)} \le \Delta < D_{(k_u)}) = P(k_l \le U \le k_u - 1) \ge 1 - \alpha$. The associated point estimator is the Hodges–Lehmann shift estimator $\hat{\Delta} = \operatorname{median}_{i,j}(Y_j - X_i)$.

Proof. $D_{(k_l)} \le \Delta \iff U(\Delta) \le mn - k_l$ and $\Delta < D_{(k_u)} \iff U(\Delta) \ge mn - k_u + 1$. Since $U(\Delta)$ has the null distribution, symmetric about $mn/2$, the probability equals $P(k_l \le U \le k_u - 1)$. ■

In Example 4.1 the six differences $Y_j - X_i$ are -0.4, 0.9, 1.1, 1.8, 2.4, 3.1, so $\hat{\Delta} = 1.45$. For large samples, $Q_U(\alpha) \approx \frac{mn}{2} - z_\alpha\sqrt{mn(N+1)/12}$.

Ranks as a probability transform. Under H_0, $U_i = F(Z_i)$ are i.i.d. uniform for all N pooled observations, and $E\,U_{(k)} = k/(N+1)$. So, up to the factor $N+1$, the ranks are the expected uniform order statistics: the rank-sum test is a permutation test applied after (implicitly) transforming the unknown F to the uniform. This is why its null distribution is the same for every continuous F, and also why it needs $F = G$ under H_0: if the null allows unequal shapes, the transform is not common to both samples.

Paired data. With pairs (X_i, Y_i), analyze the differences $Y_i - X_i$ with the one-sample methods of Chapter 2.

4.3 Tests for scale

Assume a common center and $G(x) = F(x/\gamma)$. Ranks alone are useless here (a scale change does not shift W), so the ranks are replaced by *scores* that are large in the tails:

- **Ansari–Bradley:** scores $1, 2, 3, \ldots$ from both ends inward. A small sum of Y-scores means Y is more dispersed.
- **Siegel–Tukey:** an alternating assignment of $1, 2, \ldots, N$ from both ends, so the null distribution equals that of W (historically convenient, now unnecessary).
- **Mood:** scores $(R - \frac{N+1}{2})^2$. A large sum of Y-scores means Y is more dispersed.

Each is a permutation test of the score sum, so it is distribution-free under $F = G$.

If the centers may differ, other statistics are available.

- Under normality, use $F = S_Y^2/S_X^2 \sim F_{n-1,m-1}$ when $\gamma = 1$. This test is very sensitive to non-normality and outliers.
- The ratio of median absolute deviations $\mathrm{MAD}(Y)/\mathrm{MAD}(X)$ is a robust alternative.
- Alternatively, use any of the score statistics after centering each sample at its own median.

With unequal centers $F = G$ is no longer the null, so for the last two statistics the null distribution is obtained by permuting the median-centered observations (mean-centered for the F ratio). This is only *approximately* valid, since the centered observations are only approximately exchangeable. Without centering, a location difference masquerades as a scale difference.

4.4 The Kolmogorov–Smirnov test

For general $H_0 : F = G$ against $F \neq G$, compare the empirical distribution functions $\hat{F}_m(x) = \frac{1}{m}\sum I(X_i \le x)$ and $\hat{G}_n$:

$$K = \sqrt{\frac{mn}{m+n}}\,\sup_x |\hat{F}_m(x) - \hat{G}_n(x)|.$$

The supremum is attained at an observation, so K is a function of the ranks only. Hence its null distribution is distribution-free (for continuous F). As $m, n \to \infty$, $P(K \le x) \to 1 - 2\sum_{j\ge1}(-1)^{j-1}e^{-2j^2x^2}$ (Kolmogorov). The KS test detects any difference (location, scale, shape), but against a pure shift it is less powerful than the rank-sum test.

Why K depends only on ranks [Supplement]. Between consecutive pooled order statistics both empirical distribution functions are constant, so the supremum is a maximum over the N pooled points. At the kth smallest pooled point, $\hat{F}_m$ and $\hat{G}_n$ equal the proportions of X's and Y's among the k smallest, which are determined by the ranks. Under H_0 the rank configuration is uniform over $\binom{N}{m}$ splits. ■

4.5 Which test?

In the shift model with $F = G$ under H_0, every permutation or rank test is exact. Pitman AREs of the rank-sum test relative to the t-test equal $12\sigma^2(\int f^2)^2$, as in the one-sample case:

F	uniform	normal	logistic	Laplace	Cauchy
ARE(Wilcoxon, t)	1	$3/\pi \approx 0.955$	$\pi^2/9 \approx 1.10$	1.5	∞

(For the exponential shift family the value is 3.) On type I error: rank and permutation tests are exact. The two-sample t-test is close to nominal for light tails and conservative for heavy-tailed or skewed populations. In small-sample power studies, the permutation t-test is slightly ahead for uniform data, the t-test for normal data, the rank-sum test for Laplace-type tails, and the difference-of-medians permutation test for Cauchy-type tails. Roughly: the t-test or the permutation t-test for light tails and small samples; the rank-sum test for moderate to heavy tails; the difference-of-medians permutation test for extremely heavy tails.

4.6 Unequal variances under the null

If H_0 is "equal locations" but the scales differ, neither the rank-sum test nor any permutation test is guaranteed to hold its level. Options are Welch's t-test (CLT-based), a variance-stabilizing transformation, or a studentized rank statistic (Fligner–Policello, Chung–Romano) that is asymptotically valid. The bootstrap tests of Chapter 9 also apply.

4.7 Selected exercises

Motivation. The exercises prove the probability-integral facts behind rank tests, explain why the Mann–Whitney count is not binomial, and derive the null moments of the Ansari–Bradley scale statistic.

Exercise 4.6 (Probability integral transform)

Let $X_1, \dots, X_N \overset{\text{i.i.d.}}{\sim} F$ with F continuous.
(a) Show that $U_i = F(X_i)$ are i.i.d. Uniform$(0, 1)$.
(b) With $U_{(1)} < \cdots < U_{(N)}$ the order statistics, show that $E\,U_{(i)} = i/(N + 1)$.

Solution.

(a)

Step 1. Fix $u \in (0, 1)$. Since $F(x) \to 0$ as $x \to -\infty$ and $F(x) \to 1$ as $x \to \infty$, the set $\{x : F(x) \le u\}$ is nonempty and bounded above. Let x_u be its supremum. Since F is nondecreasing, $F(x) \le u$ for all $x < x_u$, and $F(x) > u$ for all $x > x_u$. Letting $x \uparrow x_u$ and using continuity gives $F(x_u) \le u$; letting $x \downarrow x_u$ gives $F(x_u) \ge u$. So $F(x_u) = u$.

Step 2. Hence $F(x) \le u$ iff $x \le x_u$. So $\{F(X) \le u\} = \{X \le x_u\}$ and $P(F(X) \le u) = F(x_u) = u$. This is the uniform distribution function.

Step 3. The U_i are functions of the independent X_i, hence independent.

(b)

Step 1. $U_{(i)} > u$ iff fewer than i of the U's are $\le u$, and $\#\{j : U_j \le u\} \sim B(N, u)$. Hence

$$P(U_{(i)} > u) = \sum_{k=0}^{i-1} \binom{N}{k} u^k (1-u)^{N-k}.$$

Step 2. For a nonnegative variable, $EV = \int_0^\infty P(V > u)\, du$. Using the Beta integral $\int_0^1 u^k(1-u)^{N-k}\, du = \frac{k!\,(N-k)!}{(N+1)!}$,

$$E\,U_{(i)} = \sum_{k=0}^{i-1} \binom{N}{k} \frac{k!\,(N-k)!}{(N+1)!} = \sum_{k=0}^{i-1} \frac{1}{N+1} = \frac{i}{N+1}.$$

Together with (a), this justifies the paragraph "Ranks as a probability transform" in Section 4.2.2. ■

Exercise 4.8 (Why $U(\Delta)$ is not binomial)

In the shift model, X_i and $Y_j - \Delta$ are independent with the same continuous distribution.
(a) Show that every indicator $I(Y_j - X_i > \Delta)$ is Bernoulli($\frac{1}{2}$).
(b) Explain why $U(\Delta) = \sum_{i,j} I(Y_j - X_i > \Delta)$ is nevertheless not $B(mn, \frac{1}{2})$, as the analogous count was in the sign test. Compute $\operatorname{Var} U(\Delta)$ and compare it with $mn/4$.

Solution. Put $Y_j' = Y_j - \Delta$. Then $X_1, \dots, X_m, Y_1', \dots, Y_n'$ are i.i.d. continuous, and $A_{ij} = I(Y_j' > X_i)$.

(a) Ties have probability 0: $P(X_i = Y_j') = \int P(X_i = y)\, dF(y) = 0$. By exchangeability of (X_i, Y_j'), $P(Y_j' > X_i) = P(X_i > Y_j')$, so each equals $\frac{1}{2}$.

(b)

Step 1. Indicators sharing an index are dependent. For $j \neq j'$,

$$P(A_{ij} = A_{ij'} = 1) = P(X_i < \min(Y_j', Y_{j'}')) = \frac{1}{3},$$

since each of three i.i.d. continuous variables is the smallest with probability $\frac{1}{3}$. So $\operatorname{Cov}(A_{ij}, A_{ij'}) = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$. Likewise $\operatorname{Cov}(A_{ij}, A_{i'j}) = P(\max(X_i, X_{i'}) < Y_j') - \frac{1}{4} = \frac{1}{12}$ for $i \neq i'$. Indicators with no common index are independent.

Step 2. There are $mn(n-1)$ ordered pairs of the first kind and $mn(m-1)$ of the second, so

$$\operatorname{Var} U(\Delta) = \frac{mn}{4} + \frac{mn(n-1) + mn(m-1)}{12} = \frac{mn(N+1)}{12}.$$

Step 3. This exceeds the binomial variance $mn/4 = 3mn/12$ whenever $N \ge 3$. Binomial quantiles would therefore give intervals that are too short. The value agrees with $\operatorname{Var}(W)$ in Section 4.2.1, as it must, since $U(\Delta)$ has the null law of $U = W - n(n+1)/2$ (Proposition 4.1). In the sign test the indicators involve different observations and are independent; here each X_i is compared with every Y_j'. ■

Exercise 4.11 (Ansari–Bradley moments)

Let $N = m+n$ be even. The Ansari–Bradley test assigns the scores $1, 2, \dots, N/2, N/2, \dots, 2, 1$ to the pooled order positions, and AB is the sum of the scores of the Y's. Find $E(AB)$ and $\operatorname{Var}(AB)$ under $H_0 : F = G$.

Solution.

Step 1. Under H_0 the Y-positions form a uniform random n-subset of $\{1,\dots,N\}$. So AB is n times the mean of a sample drawn without replacement from the finite population of scores, in which each value $1,\dots,N/2$ occurs twice.

Step 2. The population mean and mean square are

$$\mu = \frac{2}{N}\sum_{k=1}^{N/2} k = \frac{N+2}{4}, \qquad \frac{2}{N}\sum_{k=1}^{N/2} k^2 = \frac{(N+2)(N+1)}{12},$$

so $\sigma^2 = \frac{(N+2)(N+1)}{12} - \frac{(N+2)^2}{16} = \frac{(N+2)(N-2)}{48}$.

Step 3. By Theorem 4.2,

$$E(AB) = n\mu = \frac{n(N+2)}{4}, \qquad \operatorname{Var}(AB) = n^2 \cdot \frac{\sigma^2}{n} \cdot \frac{N-n}{N-1} = \frac{mn(N+2)(N-2)}{48(N-1)}.$$

Check: for $N = 2$ the scores are $1, 1$, $AB \equiv 1$, and indeed $E(AB) = 1$ and $\operatorname{Var}(AB) = 0$. ■